

MedEST-Open: A Weakly Supervised Benchmark for Evidence-Conditioned Medical Epistemic State Tracking

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Abstract

Clinical case narratives describe not only medical facts, but an evolving evidential process: diagnoses are introduced, ruled out, treated, improved, resolved, or recur. Existing clinical natural language processing (NLP) has studied entity extraction, assertion detection, factuality, uncertainty, and temporal relations, but proposition-level state transitions are rarely treated as the central prediction target. We introduce Medical Epistemic State Tracking (MedEST), a task in which a model predicts how new textual evidence updates the state of a medical proposition. We present MedEST-Open, a weakly supervised open benchmark prototype derived from MACCROBAT case-report text and annotations. The benchmark contains 4,207 filtered seven-way transition examples under a case-grouped split. A fine-tuned PubMedBERT sequence classifier achieves 0.9061 accuracy, 0.9075 macro F1, and 0.9062 weighted F1, outperforming a frozen PubMedBERT embedding plus balanced logistic-regression baseline by 0.2434 absolute macro F1. We frame the result carefully: the benchmark demonstrates that evidence-conditioned medical transition labels are highly learnable under a transparent weak-label protocol, but it is not an expert-gold clinical resource or a clinical decision-support system. MedEST-Open is intended as a reproducible scaffold for expert-reviewed proposition trajectory annotation and larger-scale open clinical narrative modeling.

Keywords: clinical NLP; biomedical NLP; weak supervision; PubMedBERT; proposition tracking; epistemic state tracking; medical case reports.

1 Introduction

Clinical narratives are dynamic. A case report may first state that pulmonary embolism was suspected, later report a negative computed tomography angiogram, and finally document that an alternative diagnosis was treated with clinical improvement. Static extraction systems can identify medical terms; temporal systems can order events; assertion systems can label local uncertainty or negation. A different question remains: *what did the new evidence do to the state of the prior medical proposition?*

We formalize this question as evidence-conditioned medical epistemic state tracking. A proposition such as *patient has pneumonia* may move through states such as introduced, treated, improving, and resolved. A different proposition such as *patient has pulmonary embolism* may move from suspected to ruled out. The model output is a structured state update rather than a free-form summary:

$$\text{old state} + \text{proposition} + \text{evidence} \rightarrow \text{transition} + \text{new state.} \quad (1)$$

This paper introduces MedEST-Open, a weakly supervised benchmark prototype for this transition-centric view of clinical NLP. The current release uses MACCROBAT case-report data as a seed,

generates transparent weak labels, performs automatic heuristic label audits, and evaluates several model families on seven transition labels: introduced, treated, ruled_out, improving, resolved, recurrent, and no_change.

The contributions are:

- We define Medical Epistemic State Tracking as evidence-conditioned state tracking over medical propositions.
- We construct MedEST-Open, a weakly supervised benchmark prototype with 4,207 filtered seven-way transition examples.
- We compare frozen embedding, neural multitask, and fine-tuned transformer baselines under a case-grouped split.
- We report 0.9075 macro F1 with fine-tuned PubMedBERT while explicitly documenting weak-label limitations and audit results.

2 Related Work

2.1 Biomedical pretrained language models

Biomedical NLP has been shaped by domain-adapted and domain-specific pretrained language models. BioBERT showed that additional biomedical pretraining improves downstream biomedical named entity recognition, relation extraction, and question answering [2]. PubMedBERT further argued for biomedical pretraining from scratch and reported strong performance across biomedical NLP benchmarks [1]. Surveys of biomedical pretrained models describe the broader shift toward specialized neural encoders for biomedical text [3]. MedEST-Open uses PubMedBERT because the task depends on biomedical semantics and clinical proposition wording.

2.2 Assertion, factuality, uncertainty, and time

Clinical assertion and factuality systems label whether local mentions are present, absent, possible, conditional, or factual. Biomedical factuality work models negation, speculation, and event certainty [4]. Clinical temporal extraction tasks such as Clinical TempEval identify events, time expressions, and temporal relations [5]. These literatures are adjacent but not identical to MedEST. A temporal extractor can say that a CT scan occurred after diagnostic suspicion; MedEST asks whether that scan ruled the suspected condition out. A mention-level assertion detector can label a phrase as negated; MedEST tracks proposition-level state changes across an ordered narrative.

2.3 Open clinical case-report resources

MedEST-Open deliberately starts from open case-report narratives rather than restricted electronic health record corpora. MACCROBAT provides 200 clinical case-report documents with brat standoff annotations and is useful as a compact seed for method development [6]. PMC Open Access is a natural future scale-up source, but article-level licensing must be preserved because PMC content is not uniformly reusable under the same terms [7, 8]. PMOA-TTS is relevant for future expansion

because it converts PubMed Open Access case reports into structured event-time timelines [9]. In MedEST, such timelines provide temporal scaffolding; they do not replace proposition state labels.

3 Task Definition

A case narrative is an ordered sequence of text segments:

$$C = \{s_1, s_2, \dots, s_T\}. \quad (2)$$

For each proposition p and segment s_t , the model predicts a transition label indicating how the segment changes the proposition state. Table 1 gives the filtered label space.

Table 1: Transition labels used in the filtered MedEST task.

Label	Meaning
introduced	proposition first enters the case narrative
treated	treatment is started or applied
ruled_out	evidence excludes the proposition
improving	condition or symptom improves
resolved	condition or symptom is absent or recovered
recurrent	prior problem returns
no_change	proposition is mentioned without a state change

The final transformer input uses an explicit natural-language template. Each example concatenates a task instruction, the old state, the normalized proposition, and the evidence sentence, then predicts the transition and new state. This template keeps the semantic update primitive visible to the model while remaining simple enough for reproducible sequence classification.

4 Benchmark Construction

4.1 Source data and weak supervision

The current benchmark is derived from MACCROBAT case-report text and annotations. The pipeline parses 200 cases, orders case segments, reads brat annotation spans, normalizes candidate medical propositions, applies cue and consistency rules, filters rare transition labels, and evaluates with a case-grouped split. Table 2 summarizes the benchmark construction process.

Weak labels are generated from transparent cue rules and state-consistency heuristics. For example, treatment phrases bias toward *treated*, explicit negative findings bias toward *ruled_out*, and recovery phrases bias toward *resolved*. These weak labels are useful for scaling and reproducibility, but they are not equivalent to expert clinical adjudication.

4.2 Audit of weak labels

We audited weak-label quality with automatic heuristic checks. In a 100-example audit, 84% were marked good, 8% bad proposition, 4% bad transition, and 4% ambiguous. In a 500-example audit,

Table 2: Benchmark construction summary. The current release is weakly supervised and inspectable; it is not expert-gold clinical annotation.

Stage	Operation	Output
Source cases	Parse MACCROBAT case-report text and brat annotations	200 cases
Segmentation	Order case sentences/segments and retain mention metadata	5,480 segments; 25,189 mentions
Weak labeling	Normalize propositions and apply cue/consistency rules	4,296 weak examples
Filtering	Retain seven stable transition labels	4,207 transition examples
Evaluation split	Group split by case identifier; held-out fraction 0.25; seed 17	1,022 held-out examples

76.0% were marked good, 10.6% ambiguous, 6.8% bad transition, and 6.6% bad proposition. Table 3 reports the audit. The audit estimates obvious noise modes; it does not replace expert review.

Table 3: Automatic heuristic weak-label audit. Percentages are shares of audited examples.

Sample	Good	Ambiguous	Bad transition	Bad proposition
100 examples	84.0	4.0	4.0	8.0
500 examples	76.0	10.6	6.8	6.6

5 Models and Experimental Protocol

5.1 Split protocol and metrics

All main experiments use a case-grouped split by `case_id`; held-out examples come from cases separated from training cases. The held-out fraction is 0.25 and the seed is 17. The primary metric is macro F1, because the transition distribution is imbalanced and rare labels should not be hidden by accuracy. We also report accuracy and weighted F1.

5.2 Baselines

The most-frequent baseline predicts *introduced* for every held-out example and establishes a low macro-F1 floor. The frozen-embedding baseline encodes each input template with PubMedBERT and trains balanced logistic regression over embeddings. The neural multitask EpiDelta head jointly predicts transition and state labels from EpiDelta-style inputs. A pure cue-rule system is not reported as an independent classifier against labels generated by the same rules; that evaluation would be circular. Instead, cue rules are treated as the supervision source and audited.

5.3 Fine-tuned PubMedBERT

The final model fine-tunes PubMedBERT-base (microsoft/BiomedNLP-PubMedBERT) for seven-way sequence classification. Training used CUDA on an NVIDIA GeForce RTX 3070 Ti Laptop GPU,

maximum sequence length 256, batch size 16, learning rate 2×10^{-5} , and 8 epochs. The best checkpoint occurred at epoch 5 by held-out macro F1.

6 Results

Table 4 and Figure 1 summarize model performance. Fine-tuned PubMedBERT achieves 0.9061 accuracy, 0.9075 macro F1, and 0.9062 weighted F1. This improves over the frozen PubMedBERT embedding plus balanced logistic-regression baseline by 0.2434 absolute macro F1.

Table 4: Model comparison on the filtered seven-way MedEST transition task.

Model	Accuracy	Macro F1	Weighted F1
Fine-tuned PubMedBERT	0.9061	0.9075	0.9062
PubMedBERT embeddings + balanced logistic regression	0.7369	0.6641	0.7387
Neural multitask EpiDelta head	0.6743	0.5796	0.6700
New-state logistic regression	0.6795	0.5562	0.6773
All-transition logistic regression	0.7072	0.4488	0.7162
Most-frequent baseline	0.4100	0.0830	0.2380

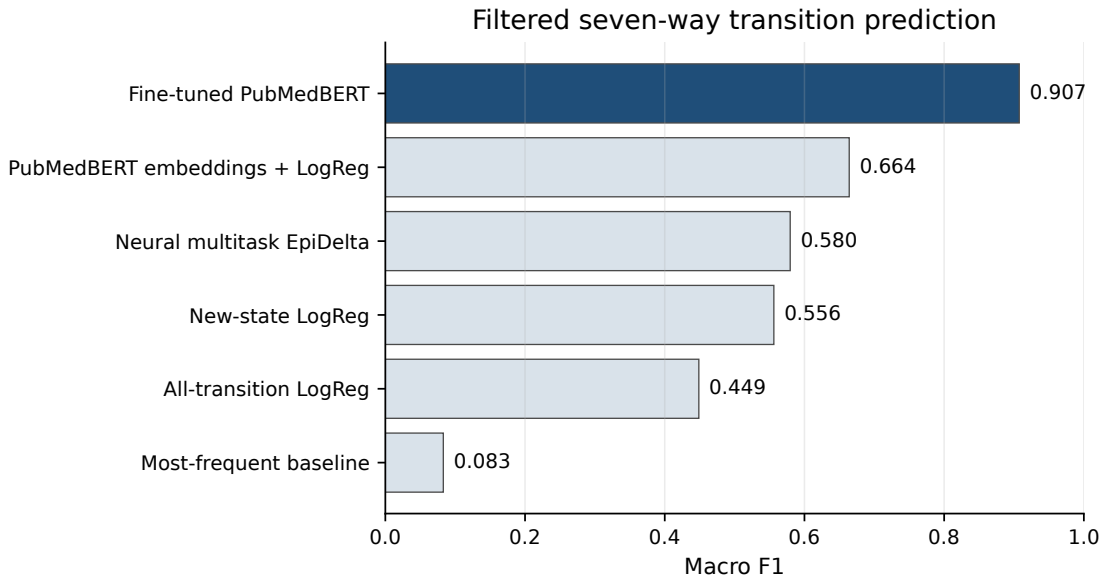


Figure 1: Model comparison on the filtered seven-way MedEST transition task.

Figure 2 shows the fine-tuning trajectory. Performance rises sharply by epoch 1 and peaks at epoch 5. Later epochs remain strong but do not improve the best macro F1, which is consistent with a small weak-label benchmark where the model rapidly learns the supervision signal.

The label distribution is shown in Figure 3. The task is imbalanced but not single-class; this motivates macro F1 as the primary metric.

Per-class results are shown in Table 5 and Figure 4. The strongest classes are *ruled_out*, *resolved*, *introduced*, and *improving*. Treated and recurrent are harder, likely because treatment and recurrence

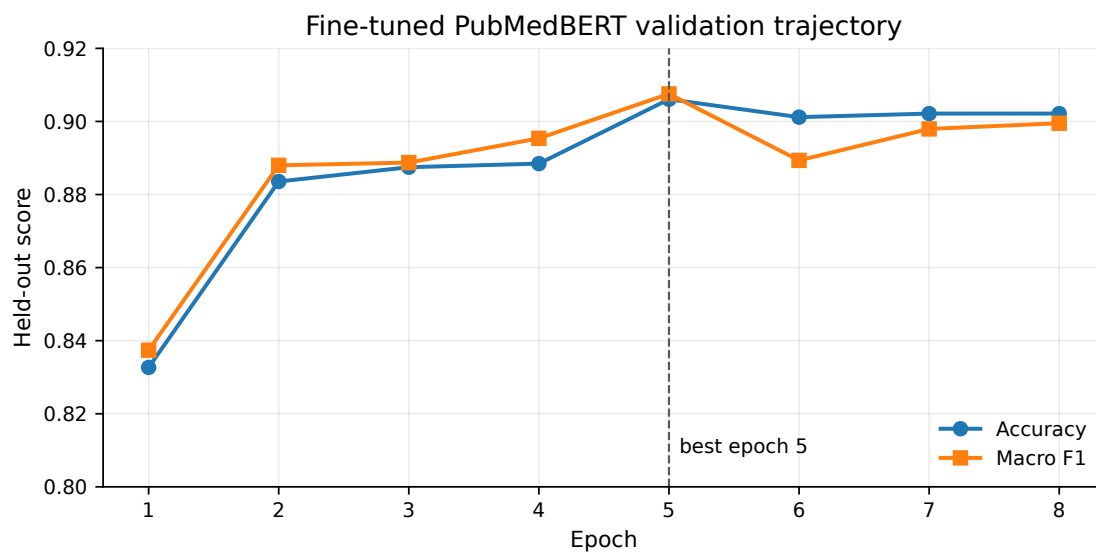


Figure 2: Fine-tuned PubMedBERT training trajectory. The best checkpoint occurs at epoch 5.

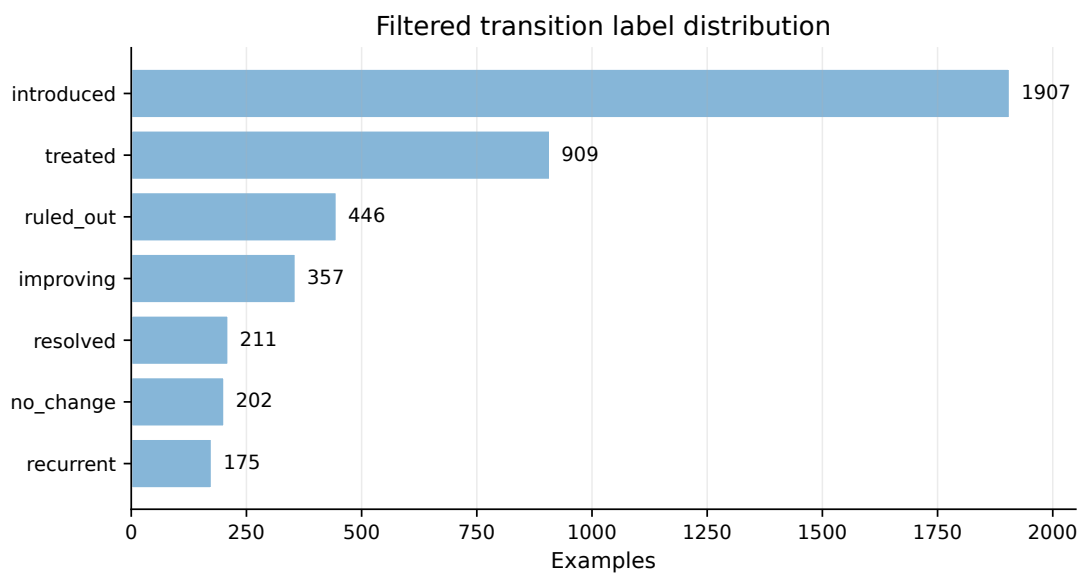


Figure 3: Filtered transition label distribution.

often require broader case context than a single local evidence segment provides.

Table 5: Fine-tuned PubMedBERT per-class transition report.

Class	Precision	Recall	F1	Support
improving	0.89	0.96	0.92	92
introduced	0.95	0.90	0.93	419
no_change	0.98	0.84	0.91	51
recurrent	0.93	0.77	0.85	53
resolved	0.90	1.00	0.95	56
ruled_out	0.94	1.00	0.97	117
treated	0.81	0.87	0.84	234

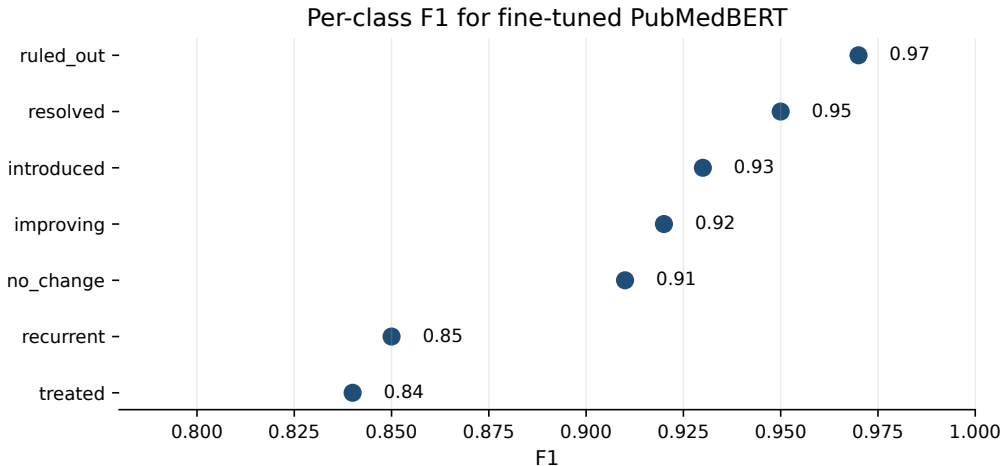


Figure 4: Per-class F1 for fine-tuned PubMedBERT.

7 Analysis and Discussion

The final result supports the central MedEST hypothesis: under a transparent weak-label protocol, evidence-conditioned transition labels are learnable from biomedical text. Fine-tuning substantially outperforms frozen embedding classification, indicating that the supervision signal is not captured fully by static sentence embeddings alone.

The result should not be overinterpreted. Because weak labels are generated from cue rules and consistency filters, a strong transformer can learn both real transition semantics and artifacts of the weak labeling function. This is still scientifically useful: it validates the operational task setup and shows that the label space is modelable. It does not establish clinical-grade reasoning or expert agreement.

The most important next step is expert-gold validation. A mature MedEST benchmark should include expert-reviewed proposition trajectories, evidence spans, agreement statistics, and saved per-example predictions for confusion-matrix analysis. The current release is best understood as a weakly supervised scaffold for that benchmark.

8 Limitations and Ethics

This work has four major limitations. First, labels are weakly supervised and heuristically audited, not expert-gold. Second, case reports are edited and publication-biased; they are not equivalent to raw hospital EHR notes. Third, the current model predicts local transition labels, not full end-to-end proposition trajectories with explicit memory and linking. Fourth, the system is a research prototype only and must not be used for clinical decision support.

Data reuse must respect article-level license terms. Future PMC Open Access expansion should preserve provenance and redistribute only compliant derived data or annotations.

9 Code, Data, and Tool Use

The project repository is available at <https://github.com/suryaavvaru/medest-open>. The current release includes code, final result documentation, weak-label audit summaries, and reproducibility notes. Raw clinical text redistribution should follow original dataset and article-license constraints.

The authors used software tools, including AI-assisted drafting and programming support, during project development and manuscript preparation. The authors remain responsible for the research design, claims, results, interpretation, and final submitted text.

10 Conclusion

MedEST-Open reframes clinical narrative understanding as evidence-conditioned state tracking over medical propositions. The current weak-label benchmark shows that fine-tuned PubMedBERT can learn seven medical transition operators with 0.9075 macro F1. The main contribution is not a clinical deployment system, but a task formulation and reproducible benchmark scaffold for proposition-level medical state updates. Expert-gold trajectory annotation is the next required milestone.

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